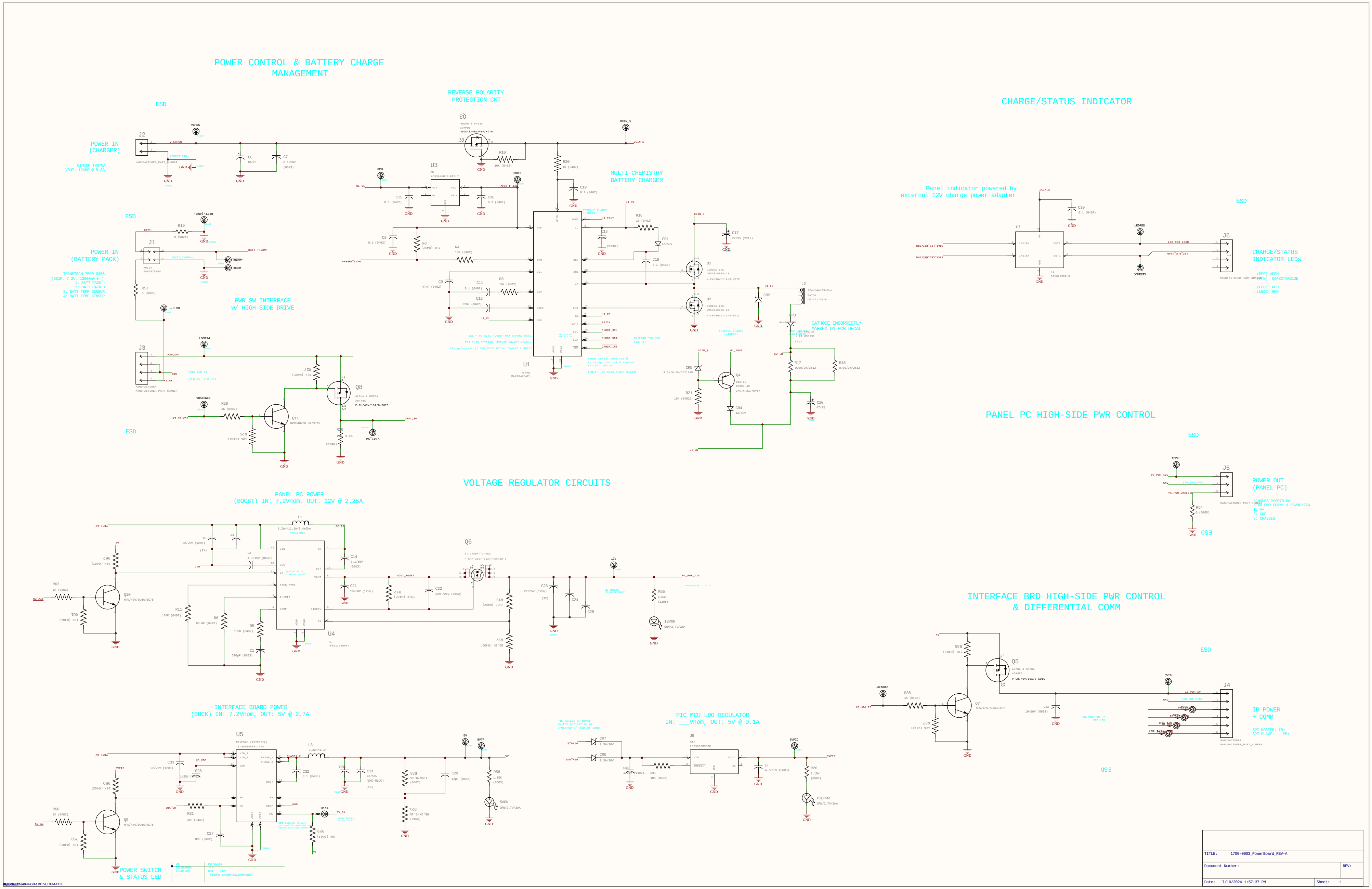

POWER CONTROL & BATTERY CHARGE MANAGEMENT
ESD
POWER IN (CHARGER)
CINCON TRH70A
VOUT: 12VDC @ 5.8A
J2
VCHRG
V_CHRGR
(CHRGR_RTN)
MANUFACTURER_PART_NUMBER
GND
C6 10/25
C7 0.1/50V (0603)
REVERSE POLARITY PROTECTION CKT
Q3 ALPHA & OMEGA AO4485 P-CH/40V/10A/8-SOIC
R10 10K (0402)
DCIN_S
R20 10 (0402)
C19 0.1 (0402)
U3 AD ADR364AUJZ-REEL7
VIN NC VOUT TRIM
U1VL
U1_VL
C15 0.1 (0402)
C16 0.1 (0402)
U1REF
REF_4.096V
MULTI-CHEMISTRY BATTERY CHARGER
ESD
BATT-(GND)
BATT-
R19 0 (0805)
POWER IN (BATTERY PACK)
TRANSTECH 7500-0356
(6S1P, 7.2V, 13000mA-hr)
1: BATT PACK -
2: BATT PACK +
3: BATT TEMP SENSOR
4: BATT TEMP SENSOR
J1 MOLEX 0462075004
(BATT_THERM-)
BATT_THERM+
THERM+
THERM-
R57 0 (0805)
BATT+
PWR SW INTERFACE w/ HIGH-SIDE DRIVE
C8 0.1 (0402)
R9 10K (0402)
R4 10K (0402)
C9 47nF (0402)
C11 0.1 (0402)
R8 10K (0402)
C12 22nF (0402)
U1 MAXIM MAX1647EAP+
DCIN IOUT VL REF THM BST DHI CCI LX CCV DACV DLO CS BATT SCL SDA INT SEL AGND PGND
TRICKLE CHARGE (LINEAR)
U1_IOUT
C13
R16 10 (0402)
CR1 1A/50V
C18 0.1 (0402)
C17 22/35 (2917)
Q1 DIODES INC. DMS3015SSS-13 N-CH/30V/11A/8-SOIC
Q2 DIODES INC. DMS3015SSS-13 N-CH/30V/11A/8-SOIC
U1_LX
L2 22uH/4A/39mOhm EATON DR127-220-R
CR2
CR3
CATHODE INCORRECTLY MARKED ON PCB DECAL
FAST CHARGE (SWITCHING)
U1_CS
R17 0.09/2W/2512
R18 0.09/2W/2512
C20 47/25
CR5 4.3V/0.8W/DO219AB
R21 10K (0402)
Q4 DIOTEC BC807-40 45V/0.8A/SOT23
CR4 1A/50V
SEL = VL SETS 4.095A MAX CHARGE RATE.
FOR THIS SETTING, RSENSE #NAME? 45mOhm.
ChargeCurrent () CMD SETS ACTUAL CHARGE CURRENT
U1_CS BATT+ CHRGR_SCL CHRGR_SDA CHRGR_INT
TO/FROM PIC MCU (SH. 2)
SMBus serial comm req'd to setup, control & monitor MAX1647 device
INT, an open-drain output
J3
PSWOUT
PSW_OUT
(STATLED_K)
(PSW_IN, LED_R+)
MANUFACTURER MANUFACTURER_PART_NUMBER
R27 10K (0402)
Q8 ALPHA & OMEGA AO4485 P-CH/40V/10A/8-SOIC
VBATSWEN
VBATSW_EN
R28 1K (0402)
R25 10K (0402)
Q11 NPN/40V/0.6A/SC75
R30 0.1%
VBAT_SW
ESD
CHARGE/STATUS INDICATOR
Panel indicator powered by external 12V charge power adapter
DCIN_S
C39 0.1 (0402)
U7 TI DRV8220DRLR
IN1/PH IN2/EN OUT1 OUT2 VM GND
STAT_LED_RED/GRN
LEDRED
LEDBLK
LED_RED_LEAD
LED_BLK_LEAD
J6
CHARGE/STATUS INDICATOR LEDs
(MFG) APEM
(P/N) Q6F3CYYRG12E
(LED1) RED
(LED2) GRN
ESD
PANEL PC HIGH-SIDE PWR CONTROL
ESD
12VTP
PC_PWR_12V
(IB_PWR_RTN)
PC_PWR_CHASSIS
J5
POWER OUT (PANEL PC)
ACNODES PC9070-HW
REAR PWR CONN: 9-36VDC/27W
1: V+
2: GND
3: CHASSIS
R54 0 (0805)
ESD
VOLTAGE REGULATOR CIRCUITS
PANEL PC POWER
(BOOST) IN: 7.2Vnom, OUT: 12V @ 2.25A
VBAT_SW
L1 1.2uH/11.1A/9.4mOhm
SRF=45MHz
L1_OUT
C5 22/25V (1206) (2X)
C3
C2 4.7/16V (0603)
C14 0.1/50V (0603)
R62 10K (0402)
R63 1K (0402)
12V_EN
Q10 NPN/40V/0.6A/SC75
R61 10K (0402)
R11 174K (0402)
R6 86.6K (0402)
R5 133K (0402)
C1 220pF (0603)
U4 TI TPS61178RNWT
VIN VCC EN FREQ_SYNC ILIMIT COMP SW BST VOUT DISDRV FB AGND PGND
enabled >1.2V
disabled < 0.4V
VOUT_BOOST
C21 10/50V (1206)
R12 133K (0402)
C22 22nF/25V (0402)
Q6 SI7145DP-T1-GE3 P-CH/-30V/-60A/PPAK/SO-8
R13 732K (0402)
R22 80.6K (0402)
C23 22/25V (1206) (3X)
C24
C25
12V
C26 REMOVED (formerly 100uF)
PC_PWR_12V
vout(actual) = 12.1V
R55 4.64K (1206)
12VON GRN/2.7V/2mA
INTERFACE BRD HIGH-SIDE PWR CONTROL & DIFFERENTIAL COMM
5V
R36 10K (0402)
Q5 ALPHA & OMEGA AO4485 P-CH/40V/10A/8-SOIC
IBPWREN
IB_PWR_EN
R38 1K (0402)
Q7 NPN/40V/0.6A/SC75
R37 10K (0402)
C43 10/10V (0603)
TO/FROM SH. 2 PIC MCU
5VIB
IB_PWR_5V
(IB_PWR_RTN)
IB_MOSI
IB_MISO
SPI_CLK
SPI_SS!
J4
MANUFACTURER MANUFACTURER_PART_NUMBER
IB POWER + COMM
SPI MASTER: IB+
SPI SLAVE: PB+
ESD
ESD
INTERFACE BOARD POWER
(BUCK) IN: 7.2Vnom, OUT: 5V @ 2.7A
U5 RENESAS (INTERSIL) ISL85005AFRZ-T7A
VIN_1 VIN_2 VDD EN SS PHASE_1 PHASE_2 BOOT FB COMP PG AGND EPAD
C33 22/25V (1206)
U5_VDD
C28 1/25V (0402)
5VPIC
R59 10K (0402)
R60 1K (0402)
5V_EN
Q9 NPN/40V/0.6A/SC75
R58 10K (0402)
R31 DNP (0402)
C27 DNP (0402)
PHASE1-2
L3 3.3uH/4.2A
C32 0.1 (0402)
C30
C31 47/50V (SMD-MLCC) (2X)
R23 499K/0.1% (0402)
R24 95.3K/0.1% (0402)
C29 12pF (0402)
5V
5VTP
R56 1.15K (0603)
5VON GRN/2.7V/2mA
5VOK
5V_OK
power status output to PIC
COMP external network optional for extended temp operational requirement
R29 10K (0402)
POWER SWITCH & STATUS LED
U5
ISL85005A
ISL85005
POPULATE

R31 - FCCM
(VISHAY CRCW040210K0FKEDC)
PIC MCU LDO REGULATOR
IN: ___Vnom, OUT: 5V @ 0.1A
PIC active on power switch activation or presence of charger power
DCIN_S
PSW_OUT
CR7 0.3A/20V
CR6 0.3A/20V
C36 1/10V (0402)
R35 10K (0402)
U6 STM LD2981CM50TR
VIN INHIBIT VOUT NC GND
C4 4.7/16V (0603)
5VPIC
R26 1.15K (0603)
PICPWR GRN/2.7V/2mA
POWER BOARD SCHEMATIC

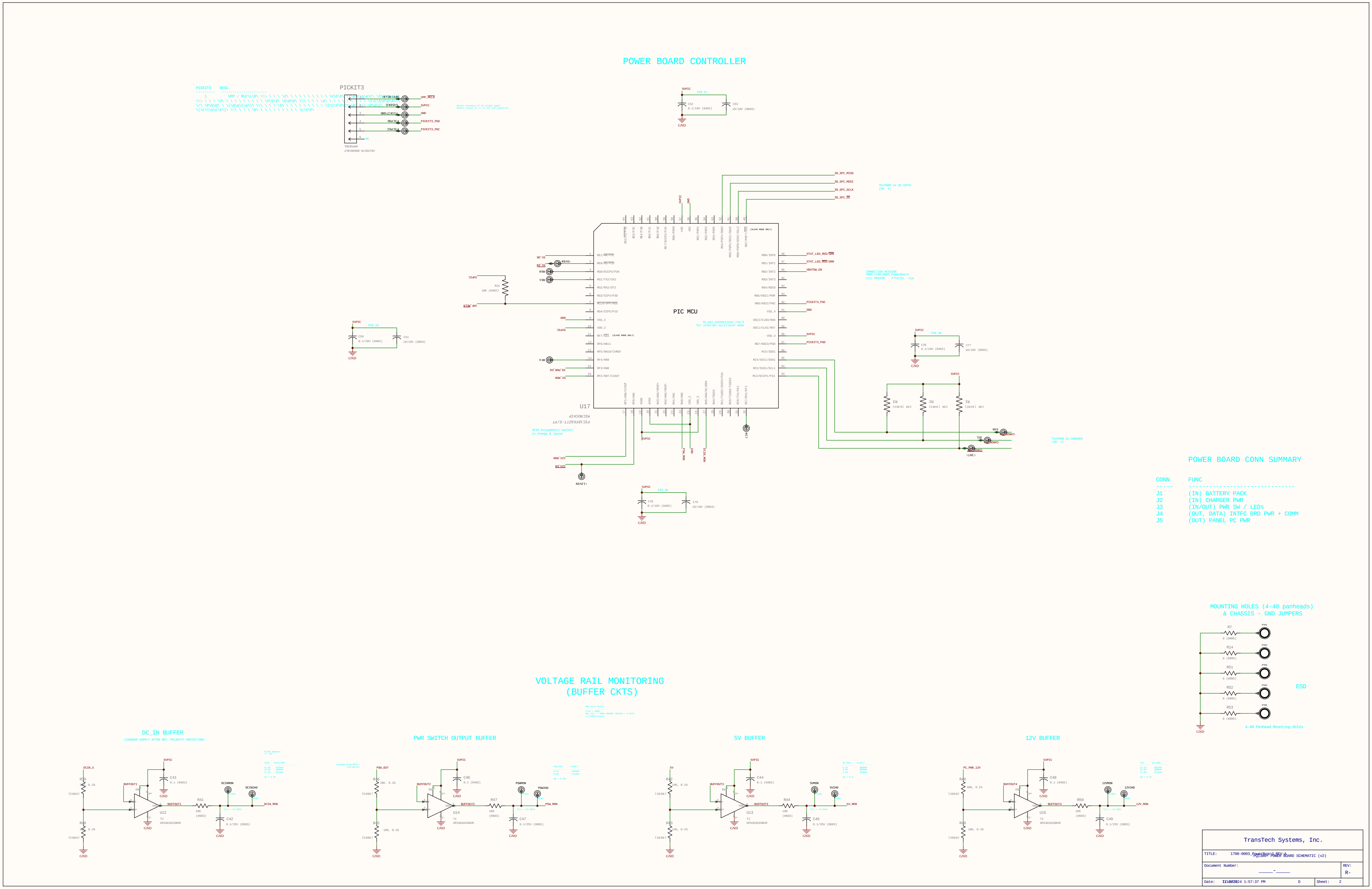
TITLE: 1700-0003_PowerBoard_REV-A

Document Number:

REV:

Date: 7/10/2024 1:57:37 PM

Sheet: 1


POWER BOARD CONTROLLER
PICKIT3
VPP_MCLR
5VPIC
GND
PICKIT3_PGD
PICKIT3_PGC
PIC MCU
U17
MICROCHIP
PIC18F6627T-E/PT
GPIO assignments subject to change @ layout
no pin connections req'd for internal oscillator mode
IB_SPI_MISO
IB_SPI_MOSI
IB_SPI_SCLK
IB_SPI_SS
TO/FROM J4 IB INTFC (SH. 1)
STAT_LED_RED/GRN
VBATSW_EN
CONNECTION MISSING FROM 1700-0003_PowerBoard (v1) DESIGN. 2/23/23, -sjp
5V_OK
5V_EN
RG0
RG1
RF4
IB_PWR_EN
5V_MON
12V_MON
12V_EN
PSW_MON
DCIN_MON
RC7
R15
10K (0402)
R3
10K (0402)
C78
0.1/16V (0402)
C79
10/10V (0603)
PIN 26
VOLTAGE RAIL MONITORING (BUFFER CKTS)
DC_IN BUFFER
(CHARGER SUPPLY AFTER REV. POLARITY PROTECTION)
PWR SWITCH OUTPUT BUFFER
5V BUFFER
12V BUFFER
DCIN_S
PSW_OUT
5V
PC_PWR_12V
BUFFOUT1
BUFFOUT2
BUFFOUT3
BUFFOUT4
U12
U13
U14
U15
TI
OPA363AIDBVR
DCINMON
DCIN2AD
PSWMON
PSW2AD
5VMON
5V2AD
12VMON
12V2AD
DCIN_MON
PSW_MON
5V_MON
12V_MON
changed from BATT+ (12/20/22)
MOUNTING HOLES (4-40 panheads) & CHASSIS - GND JUMPERS
R7
R14
R51
R52
R53
0 (0805)
PTH1
PTH2
PTH3
PTH4
PTH5
ESD
4-40 Panhead Mounting Holes

POWER BOARD CONN SUMMARY

CONN	FUNC
J1	(IN) BATTERY PACK
J2	(IN) CHARGER PWR
J3	(IN/OUT) PWR SW / LEDs
J4	(OUT, DATA) INTFC BRD PWR + COMM
J5	(OUT) PANEL PC PWR

TransTech Systems, Inc.

TITLE: 1700-0003_PowerBoard_REV-A PQI380+ POWER BOARD SCHEMATIC (v2)

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Sheet: 2